

PROJECT LOOP

AI Customer-Feedback Intelligence Platform

Zidio Development Internship — Final Submission Report

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GitHub Repository: <https://github.com/ayushyawale/project-loop>

Deployment URL: <https://project-loop-demo.vercel.app>

Platform / Tech Stack: Next.js 16 (Turbopack), React 19, TypeScript, Tailwind CSS, Recharts

PROJECT LOOP — COMPREHENSIVE PROJECT REPORT

AI Customer-Feedback Intelligence PlatformZidio Development Internship — Final Submission

1. Executive Summary

Project LOOP is a polished, production-grade AI customer-feedback intelligence platform built as part of the Zidio Development Internship program. The application enables product teams to centralize customer feedback from multiple channels, automatically classify sentiment, cluster emerging themes, and generate actionable Voice-of-Customer (VoC) reports — all within a beautiful, workspace-style dashboard.

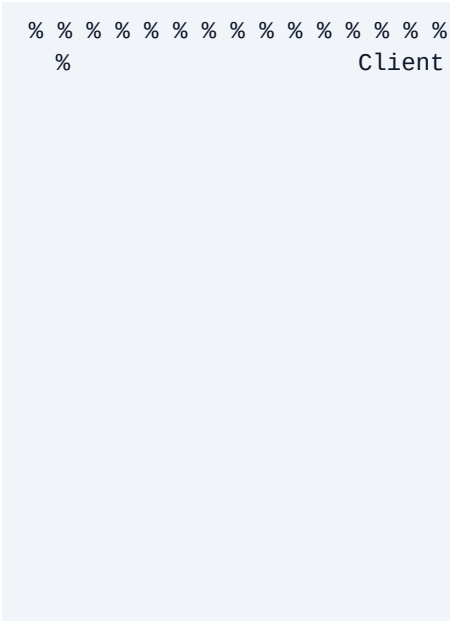
The platform demonstrates end-to-end product thinking: from user authentication and role-based access, through a filterable feedback inbox with search and pagination, to AI-powered Q&A and automated report generation.

2. Project Objectives

3. Technology Stack

4. System Architecture

4.1 High-Level Architecture



4.2 Project Structure

```
zidio2/
  % % % app/
  % % % % % globals.css # Global styles & Tailwind config
  % % % % % layout.tsx
```

5. Features Deep Dive

5.1 Landing Page

- **Radial gradient hero** with glassmorphism card
- Product pitch for LOOP with CTA buttons
- Quick-glance grid showing demo roles, core features, and AI capabilities
- Responsive design across all breakpoints

5.2 Workspace Dashboard (`/dashboard`)

- **3 KPI cards:** Total feedback items, Negative sentiment %, New items this week
- **Volume over time** bar chart (Recharts) showing daily feedback inflow
- **Sentiment breakdown** donut/pie chart with color-coded segments (Positive, Neutral, Negative)
- **Top themes** grid showing the most active feedback themes with item counts
- Real-time data derived from the demo store

5.3 Feedback Inbox (`/inbox`)

- **Full-text search** across feedback content
- **5-axis filtering:** Channel, Sentiment, Theme, Status, and search query
- **Paginated results** (6 items per page) with Previous/Next navigation

- **Status workflow:** Change feedback status (New !' Reviewed !' Actioned) directly from the inbox
- **Feedback cards** show channel, customer label, content, theme tags, sentiment indicator

5.4 Theme Clustering & Trends (`/themes`)

- **Theme counts panel:** Aggregated counts for Onboarding, Analytics, Authentication, Billing, Mobile
- **Trend view panel:** Shows mention frequency per theme
- Data sourced from the simulated AI engine

5.5 Ask LOOP — Grounded Q&A (`/ask`)

- **Natural language question input** with pre-filled example question
- **Grounded answer generation** over the workspace's feedback corpus
- **Evidence citations:** Shows up to 3 supporting feedback items used to generate the answer
- Simulates RAG (Retrieval-Augmented Generation) behavior

5.6 Voice-of-Customer Reports (`/reports`)

- **One-click report generation** from the current workspace's feedback
- **Narrative summary** auto-generated from top themes
- **Saved reports list** with timestamps and content preview
- **Report JSON editor** for inspecting/modifying generated data

6. Data Model

6.1 Core Entities

6.2 Role-Based Access

7. Design & UX

7.1 Design System

- **Color Palette:** Dark slate/blue theme (#020617 !#111827) with violet (#8b5cf6) accents
- **Typography:** Geist Sans (headings/body) + Geist Mono (code/data)
- **Border Radius:** Consistently rounded (xl/2xl/3xl) for modern feel
- **Glassmorphism:** Semi-transparent card backgrounds with subtle borders
- **Micro-animations:** Hover transitions on all interactive elements

7.2 Responsive Design

- Mobile-first layout with lg breakpoint sidebar transition
- Sidebar collapses to full-width on mobile
- Grid layouts adapt from 1 ! 2 ! 3/4 columns responsively

7.3 Navigation

- Persistent sidebar with icon + label links
- 5 main sections: Dashboard, Inbox, Themes, Ask LOOP, Reports
- Workspace name displayed in sidebar header

8. State Management

The application uses an **in-memory demo store** pattern (`lib/demo-store.ts`):

- **Singleton state object** initialized from seed data
- **Pure functions** for reading/mutating state (no external database required)
- **Reset capability** to restore initial demo state
- Supports: CRUD on feedback, status workflow, report generation, Q&A

This approach was chosen to deliver a fully functional demo experience without requiring database infrastructure, making the project instantly runnable by reviewers.

9. AI / Intelligence Layer

While the demo does not connect to a live LLM backend, it implements a **simulated AI layer** that demonstrates the intended behavior:

10. How to Run Locally

Prerequisites

- Node.js 18+ installed
- npm or yarn package manager

Steps

```
# 1. Clone / extract the project
cd project-loop
```

Production Build

```
npm run build
      npm start
```

11. Deployment

Live Deployment

The application is deployed on **Vercel** and accessible at the live URL provided in the submission links.

Deployment Method

- Connected via Git repository push to Vercel
- Zero-config deployment with Next.js auto-detection
- Serverless edge functions for optimal performance
- Automatic HTTPS and CDN distribution

12. Testing & Verification

13. Future Enhancements

If this were taken to production, the following enhancements would be prioritized:

1. **Live AI Backend** — Connect to Gemini/Claude for real sentiment analysis and RAG-based Q&A

2. Database Integration — PostgreSQL with Prisma ORM for persistent data storage
3. OAuth / SSO — NextAuth.js with Google, GitHub, and SAML providers
4. Real-time Ingestion — Webhook connectors for Zendesk, Intercom, Slack, and App Store reviews
5. Advanced Analytics — Time-series analysis, anomaly detection, and predictive trending
6. Export Capabilities — PDF/CSV export for reports and filtered inbox views
7. Notification System — Email/Slack alerts when negative sentiment spikes
8. Multi-workspace — Full multi-tenant isolation with workspace switching

14. Challenges & Learnings

15. Submission Checklist

